L13 - Quick Sort

Monday, April 27, 2020 8:35 AM

General Method

Algorithm DAC(P)
{
    if Small(P)
        return DS(P)
    else {
        divide P into smaller instances $P_1, P_2, \ldots, P_k$, K >= 1; // Divide
        Apply DAC to each of these subproblems; // Conquer
        return Combine(DAC(P_1), DAC(P_2),…, DAC(P_k)); // Combine

    //This above step ("Combine") may not be present in some of the problems that fall under this strategy. Note, return will always be executed.
    }
}

quickSort(Arr, l, r)
1. if l < r {
2.   let p = partition(Arr, l, r)
3.   quickSort(Arr, l, p-1)
4.   quickSort(Arr, p+1, r)

Arr[l..i] partition that contains elements <= PIVOT
Arr[i+1..j-1] partition that contains elements > PIVOT
Arr[j..r-1] elements that the Partition algorithm has not yet processed

Example of Partition: right-most element is chosen as PIVOT

i	l, j							r
	12	18	17	11	13	15	16	14

i, l	j						r
12	18	17	11	13	15	16	14

i, l		j					r
12	18	17	11	13	15	16	14

i, l			j				r
12	18	17	11	13	15	16	14

l	i			j			r
12	11	17	18	13	15	16	14

l		i			j		r

12	11	13	18	17	15	16	14

l		i				j	r
12	11	13	18	17	15	16	14

l		i					j,r
12	11	13	18	17	15	16	14

l		i					r
12	11	13	14	17	15	16	18

```
PARTITION(Arr, l, r)
let pivot = Arr[r]
let i = l - 1
for j = l to r - 1 {
    if Arr[j] <= pivot {
        i = i + 1
        exchange Arr[i], Arr[j])
    }
}
exchange(Arr[i+1], Arr[r])
return i + 1
```

Example 2 of Partition:

i	l, j							r
	10	19	15	12	23	25	36	40

i, l		j					r
10	12	15	19	23	25	36	40

l	i	j					r
10	12	15	19	23	25	36	40

l		i	j				r
10	12	15	19	23	25	36	40

l			i	j			r
10	12	15	19	23	25	36	40

l				i	j		r
10	12	15	19	23	25	36	40

l					i	j	r
10	12	15	19	23	25	36	40

l						i	j, r
10	12	15	19	23	25	36	40